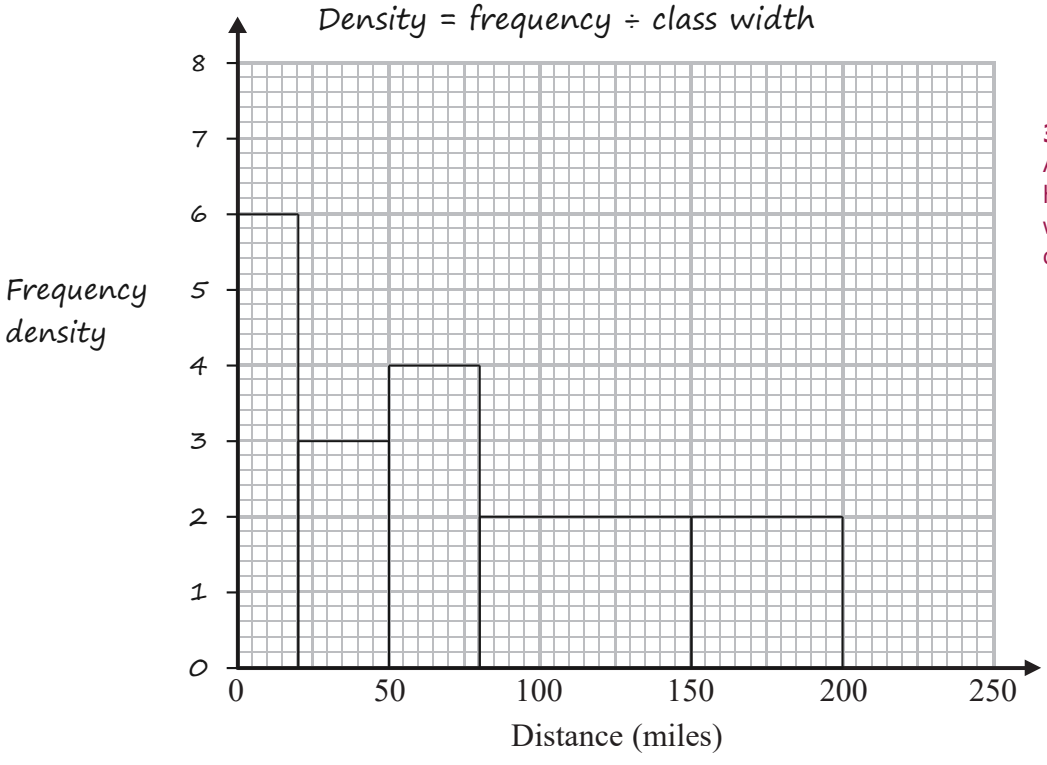
17 The table shows information about the distances 570 students travelled to a university open day.

Distance (d miles)	Frequency	Width	Density
$0 < d \leqslant 20$	120	20	6
$20 < d \leqslant 50$	90	30	3
$50 < d \leqslant 80$	120	30	4
$80 < d \leqslant 150$	140	70	2
$150 < d \leqslant 200$	100	50	2

(a) Draw a histogram for the information in the table.

Density = frequency ÷ class width



3 marks total
All five bars have correct widths and densities.

(3)

(b) Estimate the median distance.

570 ÷ 2 = 285th. Cumulative: 120, 210, 330...

Median ≈ 50 + (285 − 210) ÷ 120 × 30 = 68.75

1 mark
Correct estimate.

68.75 ..... miles

1 mark
The median lies in the third class; interpolate within it.

(2)

(Total for Question 17 is 5 marks)